

14/08

Page :

Date :

MODULE 2ALGEBRAIC STRUCTURES

Let A be any set. A mapping $f : A \times A \times \dots \times A \rightarrow A$ or $f : A^n \rightarrow A$ is a n -nary operation & ' n ' is called order of the operation.

unary: $f : A^1 \rightarrow A$; binary: $f : A^2 \rightarrow A$

* ALGEBRAIC SYSTEM

A set together with a no. of operations (binary) on the set is called an algebraic system.

abelian group

Semigroup

(i) Closure

(ii) Associative

(iii) Identity

(iv) Inverse

(v) Commutative

monoid

group

$$\forall a, b \in G ; a * b \in G$$

$$\forall a, b, c \in G ; a * (b * c) = (a * b) * c$$

$$\forall a \in G, \exists e \in G ; a * e = e * a = a$$

$$\forall a \in G, \exists a^{-1} \in G ; a * a^{-1} = e$$

$$\forall a, b \in G ; a * b = b * a$$

16/8

Ex.

1.

Let G be a set of all rational no.s except for 1 and $*$ defined as $a * b = a + b - ab$, then $(G, *)$ forms an abelian group.

Proof:

$$G = \mathbb{Q} - \{1\}$$

(i) Closure $\forall a, b \in G$

$$TP: a * b \in G$$

$$a * b = a + b - ab \in G$$

$\therefore$ closure axiom is true.

(ii) Associative $\forall a, b, c \in G$

$$TP: a * (b * c) = (a * b) * c$$

LHS:

$$\begin{aligned} & a * (b * c) \\ &= a * (b + c - bc) \\ &= a + b + c - bc - ab - ac + abc \end{aligned}$$

RHS:

$$\begin{aligned} & (a * b) * c \\ &= (a + b - ab) * c = a + b + c - ab - bc - ca + abc = LHS \end{aligned}$$

$\therefore$ associative axiom is true.

(iii) Identity $\forall a \in G, \exists e \in G$ such that:

$$a * e = e * a = a$$

LHS:

$$a * e = a$$

$$a + e - ae = a$$

$$e(1-a) = 0$$

$$e = 0 \in G$$

$\therefore$ identity axiom is true.

(iv) Inverse $\forall a \in G, \exists a^{-1} \in G$ such that,

$$a * a^{-1} = a^{-1} * a = e = 0$$

LHS:

$$a * a^{-1} = 0$$

$$a + a^{-1} - aa^{-1} = 0$$

$$a^{-1}(1-a) = -a$$

$$a^{-1} = \frac{-a}{1-a} = \frac{a}{a-1} \in G$$

$\therefore$ inverse axiom is true.

(v) Commutative $\forall a \in G, \forall b \in G$

$$TP: a * b = b * a$$

$$LHS: a * b$$

$$= a + b - ab = b + a - ba$$

$$= b * a = RHS$$

$\therefore$ commutative axiom is true.

$\therefore (G, *)$ is an abelian group

2.

If M_2 is a set of all 2 by 2 non singular matrices, check whether M_2 forms a group under matrix mul.

$$M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix}, ad - bc \neq 0, \forall a, b, c, d \in \mathbb{R} \right\}$$

(i) Closure $\forall A, B \in M_2$

$$AB \in M_2$$

$\therefore$ closure axiom is true

(ii) Associative $\forall A, B, C \in M_2$

$$\text{TPT: } (AB) \cdot C = A \cdot (BC)$$

$\therefore$ associative axiom is true

(iii) Identity $\forall A \in M_2, \exists I \in M_2$

$$\text{TP: } AI = IA = A \text{ then}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in M_2$$

$\therefore$ identity axiom is true.

(iv) Inverse $\forall A, A^{-1} \in M_2$ st,

$$AA^{-1} = A^{-1}A = I$$

$$A^{-1} = \frac{1}{|A|} (\text{adj } A) \in M_2.$$

$\therefore$ inverse axiom is true

Hence $(M_2, \text{matrix mul})$ is a group.

3.

$*$ on $\mathbb{Q}^+$

$a * b = \frac{ab}{3}$, $\text{ST } (\mathbb{Q}^+, *)$ is an abelian group

(i) Closure $\forall a, b \in \mathbb{Q}^+$

$$a * b = \frac{ab}{3} \in \mathbb{Q}^+$$

$\therefore$ closure axiom is true.

(ii) Associative $\forall a, b, c \in \mathbb{Q}^+$

$$\text{TP: } a * (b * c) = (a * b) * c$$

$$\text{LHS: } a * (b * c)$$

$$= a * \frac{bc}{3} = \frac{abc}{9} \in \mathbb{Q}^+$$

$$\text{RHS: } (a * b) * c$$

$$= \frac{ab}{3} * c = \frac{abc}{9} \in \mathbb{Q}^+ = \text{LHS}$$

$\therefore$ associative axiom is true.

16/8

(ciii) Identity $\forall a \in \mathbb{Q}^+, \exists e \in \mathbb{Q}^+$

TP: $a * e = e * a = a$

LHS: $a * e = a$

$$\frac{ae}{3} = a$$

$$e = 3 \in \mathbb{Q}^+$$

 $\therefore$ identity axiom is true.

(iv) Inverse $\forall a, a^{-1} \in \mathbb{Q}^+$

TP: $a * a^{-1} = a^{-1} * a = e$

$$a * a^{-1} = e$$

$$\frac{aa^{-1}}{3} = e$$

$$aa^{-1} = 9$$

$$a^{-1} = \frac{9}{a} \in \mathbb{Q}^+$$

 $\therefore$ inverse axiom is true.

(v) Commutative $\forall a, b \in \mathbb{Q}^+$

TP: $a * b = b * a$

 $\therefore (\mathbb{Q}^+, *)$ is abelian group

$$a * b = \frac{ab}{3} = \frac{ba}{3}$$

$$= b * a$$

 $\therefore$ commutative axiom is true.

Ex.

1.

$$G = \{0, 1, 2, \dots, 5\}$$

check whether $\langle G, +_8 \rangle$ is a group.

- (i) Closure
 - (ii) Associative
 - (iii) Identity
 - (iv) Inverse
- } for group

$+_8$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	6
2	2	3	4	5	6	7
3	3	4	5	6	7	0
4	4	5	6	7	0	1
5	5	6	7	0	1	2

(i) closure $\forall a, b$

$$a +_8 b \in G$$

$$5 +_8 1 = 6 \notin \{0, 1, 2, 3, 4, 5\}$$

$\therefore$ closure axiom is not satisfied.

$\therefore \langle G, +_8 \rangle$ is not a group.

2. $\langle G, +_4 \rangle$; $G = \{0, 1, \dots, 5\}$

$+_4$	0	1	2	3	4	5
0	0	1	2	3	0	1
1	1	2	3	0	1	2
2	2	3	0	1	2	3
3	3	0	1	2	3	0
4	0	1	2	3	0	1
5	1	2	3	0	1	2

(i) Closure $\forall a, b \in G$

$$a +_4 b \in G$$

$$2 +_4 4 = 2 \in G$$

$\therefore$ closure axiom is true //

(ii) Associative $\forall a, b, c \in G$

$$a +_4 (b +_4 c) = (a +_4 b) +_4 c$$

$\therefore$ associative axiom is true //

(iii) Identity $\forall a \in G$

$$0 +_4 \{0, 4\} = 0$$

2

3

4 +

$\therefore$ Since $4 \in G$ does not have an identity element, identity axiom fails.

22/8

Page :
Date :

$+$	1	4	5
1	2	1	2
4	1	0	1
5	2	1	2

$$A \subseteq G$$

$$A = \{1, 4, 5\}$$

i) closure axiom fails $\therefore (A, +)$ not a group.

$$1 + 1 = 2 \text{ which is not part of } A.$$

22/8 * SUB SEMI GROUP $\rightarrow$ only closure

Let $(S, *)$ be a semi group & $T \subseteq S$. If T is closed under $*$ then $(T, *)$ is called sub semi group.

* SUB MONOID $\rightarrow$ closure and identity

Let $(M, *, e)$ be a monoid & $T \subseteq M$. If T is closed under $*$ & $e \in T$ then $(T, *, e)$ is called sub monoid.

23/8 * IDEMPOTENT ELEMENT

$a \in G$, is called an idempotent element if it satisfies, $a * a = a$, where $(G, *)$ is the set.

* THEOREM I

For any commutative monoid $(M, *)$, the set of all idempotent elements of M forms a sub-monoid.

Proof:

Let $(M, *)$ be a commutative monoid.

$$M_1 = \{ a \in M \mid a * a = a \}$$

TPT: M_1 forms a sub-monoid

i) closure $\forall a, b \in M_1$.

$$\text{TP: } a * b \in M_1 \Rightarrow a * a = a \quad \text{i.e. TP: } (a * b) * (a * b) = a * b$$

$$b * b = b$$

LHS:

$$(a * b) * (a * b)$$

$$= (a * b) * (b * a) = (a * a) * (b * b)$$

$$= a * b = \text{RHS}$$

$\therefore$ closure axiom is true.

ii) Identity $\forall e \in M_1$.

Since $e \in M_1$ is identity element, $e * e = e$

$\therefore$ identity axiom is true.

$\therefore M_1$ forms a sub-monoid.

NOTE:

For $(\mathbb{Z}, \cdot)$ the idempotent set is $\{0, 1\}$.

For $(\mathbb{Z}, +)$ the idempotent set is $\{0\}$.

* CYCLIC GROUP

A group $(G, *)$ is called a cyclic group if there exists $a \in G$ such that $x = a^n$ where $n \in \mathbb{Z}^+$.

eg:

$$G = \{1, -1, i, -i\}$$

cyclic
base generator

for varying values of i^n where $n \in \mathbb{Z}^+$, the elements of G are generated.

• PROPERTIES:

Every cyclic group is an abelian group.

Proof:

$$\forall a, b \in G / a = x^m, b = x^n \quad m, n \in \mathbb{Z}^+$$

$$\text{TP: } a * b = b * a$$

Commutative axiom:

$$\begin{aligned} a * b &= x^m \cdot x^n \\ &= x^{m+n} \end{aligned}$$

$$b * a = x^n \cdot x^m$$

$$= x^{m+n}$$

$\therefore$ commutative axiom is true.

$\therefore$ cyclic groups are abelian groups.

* SUB GROUP

If $(G, *)$ is a group & $H \subseteq G$ is called a sub group of G if it satisfies closure under $*$, the identity element $e \in H$ ($e \in G$). For any $a \in G$ then $\exists a^{-1} \in H$.

NOTE:

$(H, *)$ is itself a group with the same identity element as $(G, *)$.

Obviously, $(e, *)$, $(G, *)$ are called trivial subgroups, other sub groups are called proper subgroups.

* GROUP HOMOMORPHISM

Let $\langle G, * \rangle$ and $\langle H, \Delta \rangle$ be two groups. A mapping $g: G \rightarrow H$ is called group homomorphism from G to H if for any $a, b \in G$

$$g(a * b) = g(a) \Delta g(b)$$

* SEMI GROUP HOMOMORPHISM

Let $\langle S, * \rangle$ and $\langle T, \Delta \rangle$ be any two semi groups. A mapping $f: S \rightarrow T$ is said to be a semi group homomorphism iff $f(a * b) = f(a) \Delta f(b)$, for all $a, b \in S$.

eg:

$$f: (\mathbb{N}, +) \rightarrow (\mathbb{Z}^+, \times)$$

TP: $f(a+b) = f(a) \times f(b)$, then homomorphism.

order $\rightarrow$ how many times the element is multiplied to get the identity element.

generator

Page:

Date:

25/8

* THEOREM II

Let $(S, *)$, (T, Δ) and $(V, \oplus)$ be semigroups and $g: S \rightarrow T$ and $h: T \rightarrow V$ be semigroup homomorphisms. Then $(h \circ g): S \rightarrow V$ is a semigroup homomorphism from $(S, *)$ to $(V, \oplus)$.

Proof:

$$(h \circ g): S \rightarrow V, \quad a, b \in S$$

$$(h \circ g)(a * b) = h(g(a * b)) \\ = h(g(a) \Delta g(b))$$

$$(h \circ g)(a * b) = h(g(a)) \oplus h(g(b))$$

$$\therefore (h \circ g)(a * b) = (h \circ g)(a) \oplus (h \circ g)(b), \text{ hence proved.}$$

Thus $h \circ g$ is a semigroup homomorphism.

* ORDER OF AN ELEMENT

Let $\langle G, * \rangle$ be a group. Let $a \in G$ with $a^n = e$, where n is the least +ve int satisfies above. Then we say n is the order of a or $o(a) = n$.

Ex:

1.

$$\text{Let } G = \{1, -1, i, -i\}$$

(i) is G a group under multiplication.

(ii) is G abelian under multiplication.

	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	1	-1
-i	-i	i	-1	1

from the table,

- (i) closure axiom is true.
- (ii) associative axiom is true.
- (iii) $e = 1 \in G$, identity axiom is true.
- (iv) inverse axiom is true.
- (v) commutative ^{axiom} is true.

$\therefore G$ under multiplication is true.

Ex.

1-20

Find all the subgroups of $\langle \mathbb{Z}_6, + \rangle$.

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

$$\langle 0 \rangle = \{0\} - \text{trivial subgroup}$$

$$\langle 1 \rangle = \{1, 2, 3, 4, 5, 0\} - \text{same as } \mathbb{Z}_6 \text{ (trivial)}$$

$$\langle 2 \rangle = \{2, 4, 0, 2, 4, 0\} = \{2, 4, 0\}$$

$$\langle 3 \rangle = \{3, 0\}$$

$$\langle 4 \rangle = \{4, 2, 0\}$$

$$\langle 5 \rangle = \{5, 4, 3, 2, 1, 0\} - \text{same as } \mathbb{Z}_6 \text{ (trivial)}$$

} proper subgroup

trivial subgroups: $\langle 0 \rangle, \langle 1 \rangle, \langle 5 \rangle$.

proper subgroups: $\langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle$.

eg: $6 = \sqrt[2]{2} \cdot 6$

proper

→ repetition (no need) Camlin

2.

Find all subgroups of $\langle \mathbb{Z}_{12}, + \rangle$.

$$\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$\langle 0 \rangle = \{0\}$$

$$\rightarrow \langle 1 \rangle = \{1, 2, 3, 4, 5, \dots, 0\}$$

$$\langle 2 \rangle = \{2, 4, 6, 8, 10, 0\}$$

$$\langle 3 \rangle = \{3, 6, 9, 0\}$$

$$\langle 4 \rangle = \{4, 8, 0\}$$

$$\rightarrow \langle 5 \rangle = \{5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 0\}$$

$$\langle 6 \rangle = \{6, 0\}$$

$$\rightarrow \langle 7 \rangle = \{7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5, 0\}$$

$$\langle 8 \rangle = \{8, 4, 0\}$$

$$\langle 9 \rangle = \{9, 6, 3, 0\}$$

$$\langle 10 \rangle = \{10, 8, 6, 4, 2, 0\}$$

$$\rightarrow \langle 11 \rangle = \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0\}$$

proper subgroups:

$$\{0, 2, 4, 6, 8, 10\}, \{0, 3, 6, 9\}, \{0, 4, 8\}, \{0, 6\}$$

* CO-SET

For any $a \in G$, the set $a * H = \{a * h / h \in H\}$ is the left coset and the set $H * a = \{h * a / h \in H\}$ is the right coset and H is a subgroup of G .

Union of distinct cosets $\equiv G$.

28/8

Ex.

1.

Find all the cosets of H , where $G = \{1, \omega, \omega^2, \omega^3\}$ and $H = \{1, \omega^2\}$ under multiplication.

$$1 \cdot H = \{1, \omega^2\}$$

$$\omega \cdot H = \{\omega, \omega^3\}$$

$$\omega^2 \cdot H = \{\omega^2, 1\}$$

$$\omega^3 \cdot H = \{\omega^3, \omega\}$$

$\therefore$ Distinct cosets are: $\{\{1, \omega^2\}, \{\omega, \omega^3\}\}$.

2.

Find the right cosets of $\{[0], [2]\}$ in $\langle \mathbb{Z}_4, + \rangle$.

$$\mathbb{Z}_4 = \{0, 1, 2, 3\}$$

$$H + [0] = \{0, 2\}$$

$$H + [1] = \{1, 3\}$$

$$H + [2] = \{2, 0\}$$

$$H + [3] = \{3, 1\}$$

$\therefore$ Distinct cosets are: $\{\{0, 2\}, \{1, 3\}\}$.

29/8

* LAGRANGE'S THEOREM

The order of each subgroup of a finite group is a divisor of the order of the group.

Proof:

Let $(G, *)$ be a ^{finite} group with order n .

29/8

Date:

let $(H, *)$ be a subgroup with order m .

TPT:

$$\frac{O(G)}{O(H)} = \frac{n}{m}$$

let $\{h_1, h_2 \dots h_m\}$ be elements of H .

Right Coset: $H * a_1 = \{h_1 * a_1, h_2 * a_1 \dots h_m * a_1\}$

w.k.t each coset has m elements & the union of all distinct cosets will be equal to G .

$$G = (H * a_1) \cup (H * a_2) \cup (H * a_3) \dots \cup (H * a_k)$$

$$O(G) = O(H * a_1) + O(H * a_2) \dots + O(H * a_k)$$

$$n = m + m \dots + m$$

$$n = k \cdot m \quad \therefore k = \frac{n}{m} = \frac{O(G)}{O(H)} //$$

Hence Proved //

* KERNEL OF HOMOMORPHISM

let $f: G \rightarrow G'$ be a group homomorphism. The set of all elements of G which are mapped into e , identity element of G' , is called kernel of f .

$$\text{ker}(f) = \{a \in G / f(a) = e\}$$

* THEOREM IV

The necessary and sufficient condition that a non empty subset H of a group G be a subgroup is $a \in H, b \in H \implies a * b^{-1} \in H$.

Proof:

Necessary Condition

Assume H is a subgroup of G , thus since H is itself a group.

we have, $a, b \in H, a, b^{-1} \in H$

$$\therefore a * b^{-1} \in H.$$

Sufficient Condition

Assume $a * b^{-1} \in H$, to show H is a group thus it is a subgroup of G .

For,

identity: let $a \in H, a^{-1} \in H$

$$a * a^{-1} \in H$$

$e \in H$, hence identity axiom is true.

inverse: let $e \in H, a^{-1} \in H$

$$a^{-1} * e \in H$$

$a^{-1} \in H$, hence inverse axiom is true.

closure: let $b \in H, b^{-1} \in H$

for $a, b \in H$, $a, b^{-1} \in H$

$$a * (b^{-1})^{-1} \in H$$

$a * b \in H$, closure axiom is true

associative:

Since $H \subseteq G$, the elements of H are also elements of G .
Thus since composition of G is associative, H is also associative.

$\therefore H$ is a group and thus H is a subgroup of G .

* GROUP CODE / CODING THEORY

If $B = \{0, 1\}$ then $B^n = \{x_1, x_2, x_3, \dots, x_n \mid x_i \in B, i=1, 2, \dots, n\}$
is a group under the binary operation of addn. modulo 2
($\oplus$), then $(B^n, \oplus)$ is called group code.

* HAMMING DISTANCE

Suppose if x & y represent binary strings $x_1 x_2 x_3 \dots x_n$
and $y_1 y_2 y_3 \dots y_n$ then the no. of positions in the strings
for which $x_i \neq y_i$ is called Hamming Distance.

eg: $x = 1011$; $y = 1101$

$$H.D = 0+1+1+0 = 2$$

Note : 30

$(B^n, \oplus)$ is a group.

* WEIGHT

The number of ones in the binary string $x \in B^n$ is called the weight of x & is denoted by $|x|$.

* MINIMUM DISTANCE

The minimum distance of a code is defined as the minimum of hamming distances b/w all pairs of encoded words in that code.

* ENCODING FUNCTION

E is a mapping, $E: B^m \rightarrow B^n$, where m is the message & n is the encoded message, where $m < n$ always.

* GENERATOR MATRIX (G)

$G = [I_m | A]$, where I_m is the identity matrix and A is chosen random matrix of dimension $m \times (n-m)$.

* PARITY CHECK MATRIX (H)

$$H = [A^T | I_{n-m}]$$

eg:

$$G = \left[\begin{array}{cc|cc} \overbrace{I_2} & \overbrace{A} \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right] \rightarrow \begin{array}{l} 2 \times 5 \\ m \quad n \end{array}$$

$$H = \left[\begin{array}{cc|cc} \overbrace{A^T} & \overbrace{I_{5-2} = I_3} \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

1/9

Note:

If w is a message $\in B^m$, then $c(w) = wG$ & the set of code words $C = E(B^m) \subseteq B^n$ where w is a vector.

03/09

Ex:

$$E: B^2 \rightarrow B^5$$

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_A \quad \underbrace{\hspace{10em}}_I$

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$E(B^2) = wG$$

$$w = B^2 = \{00, 01, 10, 11\}$$

$$E(00) = [0 \ 0] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [0 \ 0 \ 0 \ 0 \ 0]$$

$$E(01) = [0 \ 1] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [0 \ 1 \ 0 \ 1 \ 1]$$

$$E(10) = [1 \ 0] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [1 \ 0 \ 1 \ 1 \ 0]$$

$$E(11) = [1 \ 1] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [0 \ 0 \ 0 \ 0 \ 0] \\ = [1 \ 1 \ 1 \ 0 \ 1]$$

$$\text{encoded} = \{00000, 01011, 10110, 11101\}$$

2.

$$E: B^2 \rightarrow B^5$$

$$H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$B^2 = \{00, 01, 10, 11\}$$

$$E(00) = [0 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \\ = [0 \ 0 \ 0 \ 0 \ 0]$$

$$E(01) = [0 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \\ = [0 \ 1 \ 0 \ 1 \ 1]$$

$$E(10) = [1 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \\ = [1 \ 0 \ 0 \ 1 \ 1]$$

3/9

Page:
Date:

$$E(11) = [1 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} \\ = [1 \ 1 \ 0 \ 0 \ 0]$$

Encoded messages are:

$$\{00000, 01011, 10011, 11000\}$$

$$H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$B^3 \rightarrow B^6$$

$$B^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

$$E(000) = [0 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \\ = [0 \ 0 \ 0 \ 0 \ 0 \ 0]$$

$$E(001) = [0 \ 0 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \\ = [0 \ 0 \ 1 \ 0 \ 1 \ 1]$$

$$E(010) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$$E(011) = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$E(100) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

$$E(101) = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$E(110) = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$E(111) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

encoded = { 000000, 001011, 010101, 011110, 100111, 101100, 110010, 111001 }

$Hr^T \rightarrow$ to decode.

NOTE: $Hr^T = [0]$ - messages are decoded correctly

Hr^T = any column of H - coded should be corrected

Hr^T = neither of above - there is no unique soln

Ex.

1.

$$G = \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{array} \right] \quad E: B^3 \rightarrow B^6$$

(i) 110101 (ii) 001111 (iii) 110001 (iv) 111111

$$H = \left[\begin{array}{cccccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$(i) r = [110101] ; r^T = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$Hr^T = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

It is decoded correctly, OG: 110₁₁.

$$(ii) r = [001111]$$

$$Hr^T = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ matches \#5}$$

Thus correct decoded msg: 001101.

original msg is 001₁₁.

iii) $r = 110001$

$$Hr^T = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

matches H_1 .

correct encoded msg: 110101
original message: 110.

(iv) 111111

$$Hr^T = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

does not match any column $\therefore$ there is no unique solution